

BUREAU OF INDIAN STANDARDS

DRAFT FOR COMMENTS ONLY

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भारतीय मानक मसौदा
**एकल उपयोग औषधि वितरण प्रणालियों की विश्वसनीयता के लिए
विशिष्टि और प्रदर्शन**

Draft Indian Standard
**Specification and Demonstration of System Reliability of Single-
Use Drug Delivery Systems**

[ICS 11.040.01]

Hospital Equipment and Surgical Disposable
Products Sectional Committee, MHD 12

Last date for comments:
19 September 2026

NATIONAL FOREWORD

(Adoption clause will be added later)

The text of ISO Standard has been approved as suitable for publication as an Indian Standard without deviations. Certain conventions are however not identical to those used in Indian Standards. Attention is particularly drawn to the following:

- Wherever the words ‘International Standard’ appear referring to this standard, they should be read as ‘Indian Standard’; and
- Comma (,) has been used as a decimal marker while in Indian Standards, the current practice is to use a point (.) as the decimal marker.

In this adopted standard, reference appears to certain International Standards for which Indian Standards also exist. The corresponding Indian Standards which are to be substituted in their respective places are listed below along with their degree of equivalence for the editions indicated:

<i>International Standard</i>	<i>Corresponding Indian Standard</i>	<i>Degree of Equivalence</i>
ISO 11608-1 Needle-based injection systems for medical use — Requirements and test methods — Part 1: Needle-based injection systems	IS 19282 (Part 1) : 2025/ISO 11608-1 : 2022 Needle-based injection systems for medical use — Requirements and test methods: Part 1 Needle-based injection systems	Identical
ISO 21649 Needle-free injection systems for medical	MHD/12/34619 Needle-free injection systems for medical use	Identical

use — Requirements and test methods	— Requirements and test methods	
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The Committee responsible for the preparation of this standard has reviewed the provisions of the following International Standards referred in this adopted standard and has decided that they are acceptable for use in conjunction with this standard:

<i>International Standard/ Other Publication</i>	<i>Title</i>
ISO 20072	Aerosol drug delivery device design verification — Requirements and test methods

Scope

This document addresses the specification and demonstration of capability and system reliability to successfully deliver a medicinal product as part of design verification and manufacture, utilizing empirical methods as well as modelling. This includes risk-based techniques to establish and demonstrate the required reliability level for all disposable single use drug delivery systems (DDSs) for which there is a single opportunity to deliver a single dose of a medicinal product.

DDSs covered by this document include, but are not limited to:

- needle-based injection systems (ISO 11608-1);
- needle-free injection systems (ISO 21649);
- aerosol drug delivery systems (ISO 20072).

This document does not cover:

- DDSs with containers that can be replaced;
- DDSs intended for dental use;
- finished needles;
- empty syringes;
- catheters;
- DDSs for multi-patient use;
- pumps [IEC 60601-2-24 (if electronic) or ISO 28620 (if non-electronic)];
- stand-alone prefilled syringes defined by ISO 11040-8.

In this document system reliability is limited to assessing functional performance of the DDS and does not address human factors or user interface considerations. These will be covered by a usability engineering program in accordance with IEC 62366-1.

NOTE DDSs not in scope of this document can still benefit from elements in this document but provisions of this document might not completely fulfil the basic safety and effectiveness of such DDSs.

The technical content of the document has not been enclosed as it is identical with the corresponding ISO standard. For details, please refer to ISO/TS 4452 : 2025 or kindly contact:

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